

Histology of Skin Lab Syllabus

Gartner, “Color Atlas and Text of Histology”, 7th edition, Chapter 11

Objectives: Identify structures listed in bold below (unless eliminated by Dr. Elul).

PHYSICAL SLIDE: Human skin, cornified, H & E

DIGITAL SLIDE: [Thick Skin1](#), [Thick Skin 2](#)

This is a section of thick skin. Thick skin is primarily located on palms of the hand or soles of the feet. View this slide without magnification for orientation. The dark purple edge is the epidermis *stratum corneum* (the fully-keratinized layer of epithelium).

IDENTIFICATIONS:

1. **Epidermis** (Atlas, pg. 227, Fig. 11-1)

Begin examining the epidermis from the layer **immediately superior** to the dermis.

A) Stratum basale (Germinativum) (pg. 227, Fig. 11-1)

Primarily composed of mitotic stem cells responsible for renewal of keratinocytes. This is a single row of cells. The atlas refers to this as stratum germinativum.

i) Stem cells (basal keratinocytes) (pg. 227, Fig. 11-1.3)

ii) Melanocytes (pg. 227, Fig. 11-1.3)

Melanocytes may not be easily distinguishable in this section.

Pigment-synthesizing cells of skin. Cell bodies are interspersed with keratinocytes of the stratum basale. Melanocytes tend to be round and their cytoplasm stains lighter than keratinocytes. These cells are derived from neural crest, not from keratinocytes.

B) Stratum spinosum (pg. 227, Fig. 11-1.2, 11-1.3)

Immediately superior to stratum basale. This is composed of several layers of cells and is the thickest layer of keratinocytes.

i) Keratinocytes (pg. 227, Fig. 11-1.3)

Cells are cube shaped near the stratum basale and become flatter towards the stratum granulosum.

ii) Sites of desmosome attachments (pg. 227, Fig. 11-1.3, unlabeled)

Numerous desmosomes remain attached between keratinocytes of the stratum spinosum. Keratinocytes shrink as a result of tissue preparation, so the remaining desmosome attachments give cells a “spiny” appearance.

C) Stratum granulosum (pg. 227, Fig. 11-1.2, 11-1.4)

Usually 2 to 6 layers of flattened cells with numerous distinct basophilic keratinohyalin granules.

i) Keratinocytes with keratohyalin granules (pg. 227, Fig. 11-1.4)

D) Stratum lucidum (pg. 227, Fig. 11-1.4, arrowheads & inset)

Only present in thick skin. This can be a few cell layers thick, but on your slide it probably will be just a single, discontinuous layer. Nuclei and organelles are

degenerated, and cytoplasm is becoming filled with keratin intermediate filaments.

i) Keratinocytes transitioning to stratum corneum

Flattened, translucent cells. They will stain considerably lighter than the stratum corneum.

E) Stratum corneum (pg. 227, Figs. 11-1.1, 11-1.2 & 11-1.4)

Layers of dead, fully keratinized cells. Stratum corneum is thick and stains intensely in this section (due to the densely packed keratin filaments). Cell outlines may be observed near stratum granulosum. NOTE: Some of you may observe variations in staining of this layer. This is a typical artifact.

i) Squames

You may observe these very flattened, individual tile like cells at the apical surface of the stratum corneum. These cells are sloughed off (desquamation).

2. Dermis

A) Papillary layer (pg. 227, Fig. 11-1.1; pg. 227, 11-2.3)

Most superficial layer of dermis, composed of a relatively loose irregular collagenous connective tissue (loose areolar ct). Includes dermal papillae (sometimes called ridges).

i) Dermal papillae (pg. 227, Fig. 11-1.2)

Finger like dermal projections responsible for the irregular, undulating dermis-epidermis border. These project from dermal ridges and are labeled “secondary dermal ridges” in the atlas.

ia) Capillary loops (pg. 227, Fig. 11-1.2)

These are present in virtually every dermal papilla.

ib) Meissner’s corpuscle (pg. 233, Fig. 11-4.3)

Light touch sensory receptors present in those few dermal papillae that do not possess capillary loops. **MOST SECTIONS DO NOT HAVE THESE.**

B) Reticular layer (pg. 225, Fig. 11-1.1; pg. 229, Fig. 11-2.2)

Deep layer of dermis, composed of dense irregular collagenous ct. **You are unlikely to see a well defined demarcation between papillary and reticular layers.**

i) Pacinian corpuscles (pg. 233, Fig. 11-4.4)

Relatively large, light staining structure which appears as a series of concentric rings. **Most sections do not have these.**

ii) Eccrine sweat glands (pg. 231, Fig. 11-3.4)

The base of the sweat gland (the secretory portion) is a highly coiled tube. Therefore a single section will slice through the tube many times. You should see multiple round profiles with a central lumen lined by cuboidal epithelium.

iiia) Secretory portion (pg. 231, Fig. 11-3.4, inset “b”)

Light staining cells; simple cuboidal epithelium

iiib) Duct (pg. 231, Fig. 11-3.4, inset “a”)

Darker than the secretory cells. Ducts are usually simple cuboidal epithelium, but they can have be stratified cuboidal. **The straight portion**

of each duct extends through the dermis papillary layer and the epidermis. The duct opens onto the skin surface.

3. Hypodermis (pg. 229, Fig. 11-2.1)

Hypodermis is deep to the dermis, and is a loose ct with adipose. This is the superficial fascia of gross anatomy. It is not part of the skin, although some structures found in the dermis are also present in the hypodermis, **such as eccrine glands.**

PHYSICAL SLIDE: Scalp, Human, H & E stain

DIGITAL SLIDE: [Thin Skin1](#); [ThinSkin2](#); [ThinSkin3](#)

This is a section of thin skin, with hair follicles. Thin skin covers most of the body.

1. Epidermis (pg. 229, Fig. 11-2.1, 11-2.2, 11-2.3)

MUCH thinner than epidermis of thick skin

A) Stratum basale (pg. 229, Fig. 11-2.3, “SGe”)

The single basal-most layer of cells labeled “SGe” in the figure.

B) Stratum spinosum (pg. 229, Fig. 11-2.3)

All of the cell layers between the stratum basale and stratum granulosum.

C) Stratum corneum (pg. 229, Fig. 11-2.1-2.3)

A few VERY thin layers of keratinized cells. Layer separation (sometimes referred to as a “basket-weave” pattern is an artifact.

2. Dermis (pg. 229, Fig. 11-2)

Find the structures and layers as described on the previous slide. In addition, find:

A) Hair follicle (pg. 229, Fig. 11-2.1, 11-3.1, 11-3.2)

Your sections probably will have several hair follicles cut in various orientations (i.e. longitudinal, cross, and oblique), and at various depths. **Follicle morphology is quite different at different depths.**

i) Hair bulb (pg. 231, Fig. 11-3.1)

The **terminal dilation** of the follicle, typically located in the hypodermis.

ii) Dermal papilla (pg. 231, Fig. 11-3.1)

Flame shaped protrusion of dermis into the hair bulb. **MAY NOT BE PRESENT.**

iii) Hair (pg. 231, Fig. 11-3.1, 11-3.2)

Composed of hard keratin.

v) External root sheath (pg. 231, Fig. 11-3.2)

Several layers of cells that surround the internal root sheath at the base of the follicle, and extend to the top of the follicle. External root sheath is continuous with epidermis.

B) Arrector pili muscle (pg. 229, Fig. 11-2.1, 11-3.2)

Strand of smooth muscle that inserts into the connective tissue sheath of the hair follicle. This muscle normally inserts near the level of the hair bulb, and extends just deep to the sebaceous gland. **THESE ARE DIFFICULT TO FIND.**

C) Sebaceous glands (pg. 229, Fig. 11-2.1, 11-3.3)

Large, light staining lobular glands that empty into follicles. In section, you will often observe these glands without their associated hair follicle, which may be out

of section. These glands are filled with light-staining cells that usually have a small, pyknotic nucleus and **a cytoplasm filled with lipid-like vacuoles.**